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Long-term cognitive deficits in memory, attention, and executive functioning in post-COVID-19 syndrome patients

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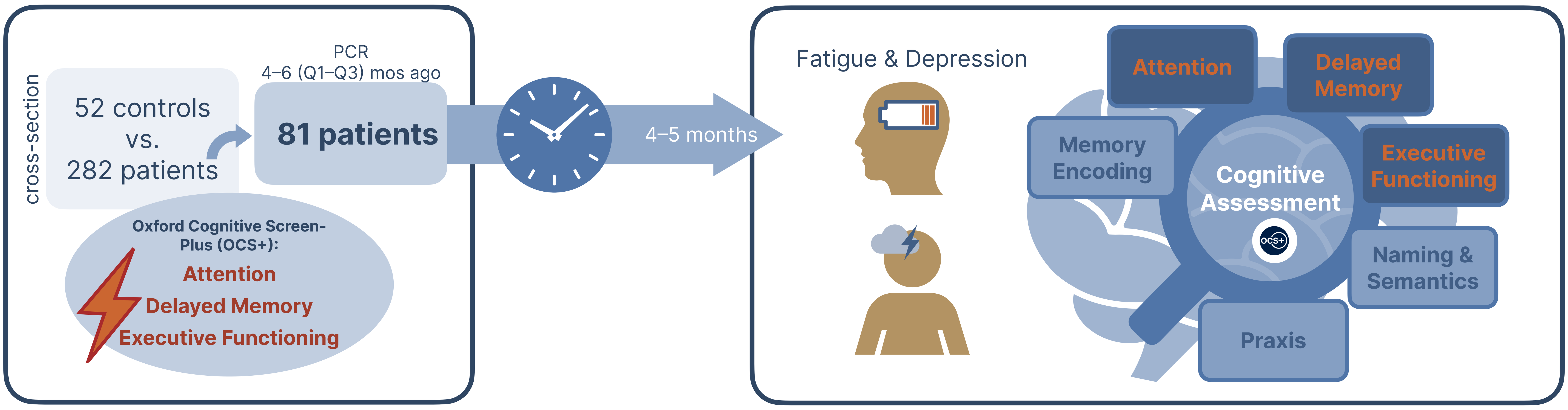
Background & Methods

- Previous study: **deficits in attention, delayed verbal memory, and executive functioning** in comparison to healthy controls (Kozik et al., 2023)
- Cross-sectional analysis: no effect of the time since infection → first indication of symptom persistence

Important question: **How do these cognitive deficits change over time?**

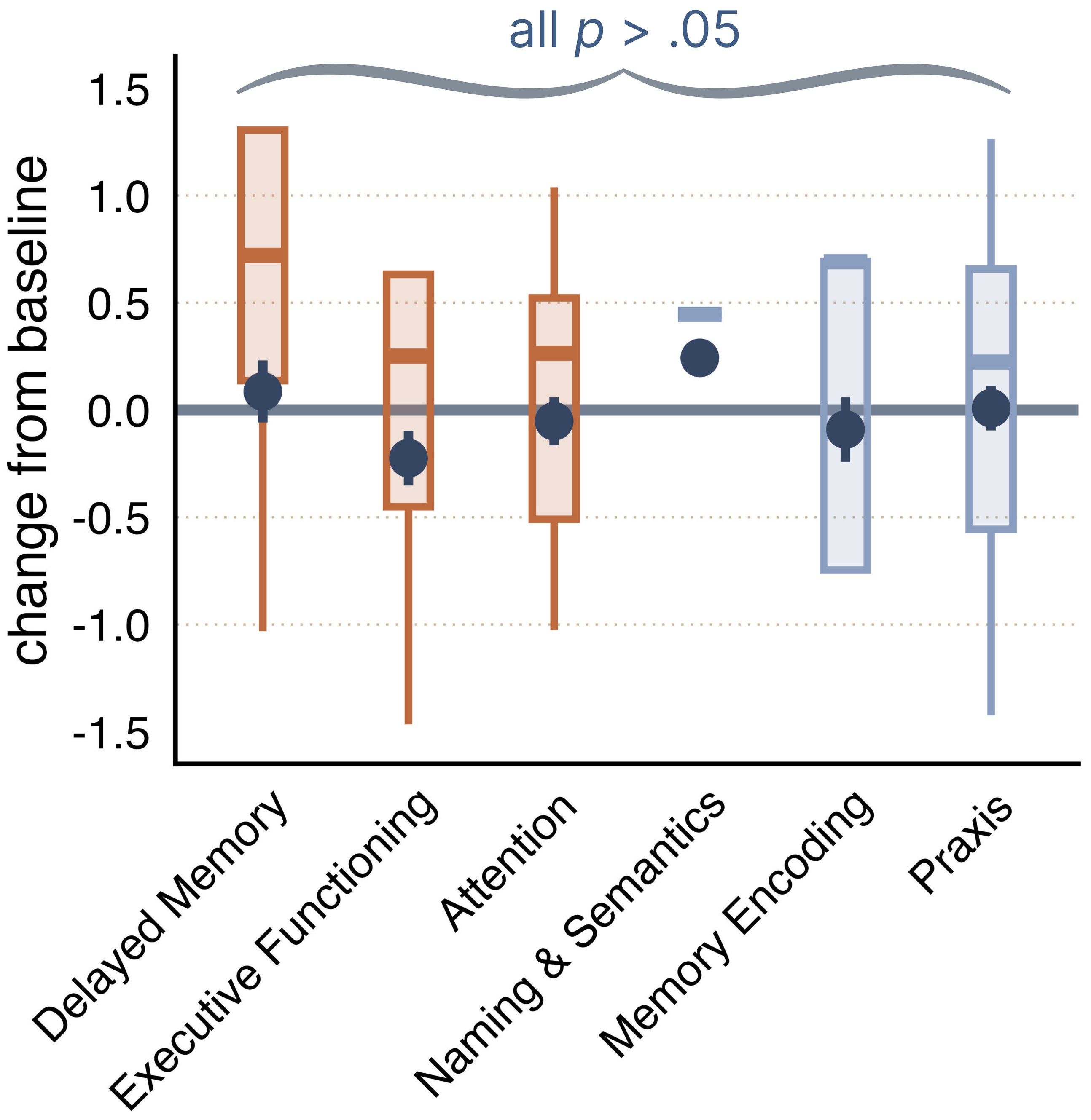
Longitudinal analysis: re-examination of **81 patients** previously assessed 5 months (median) after infection



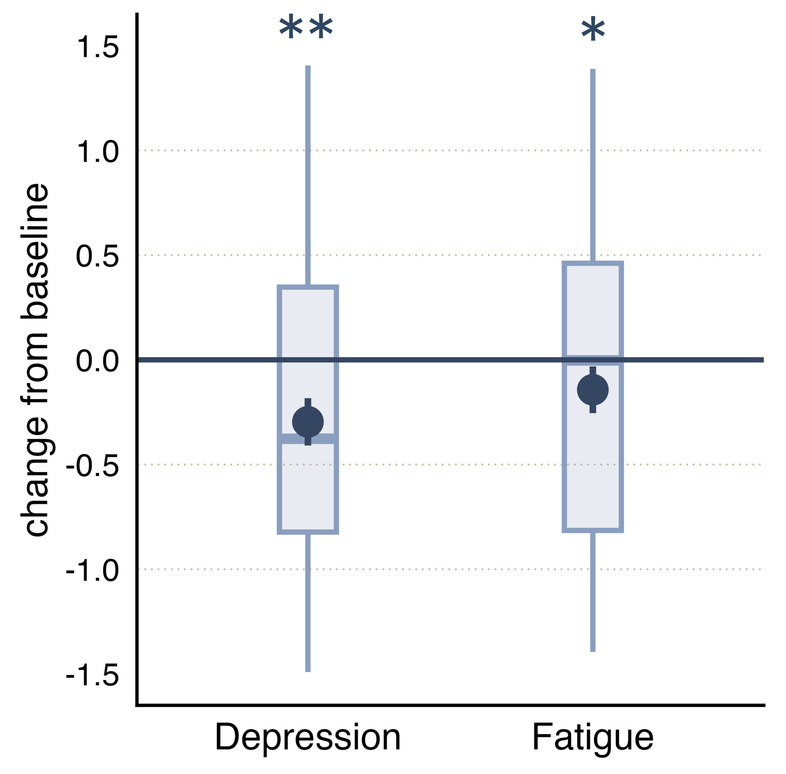
Results

Cognitive Assessment

- **No observed changes in any cognitive domain** (all $p > .05$)
- Small positive tendency (n.s.) in delayed verbal memory ($V = 820$, $HL = .5$, 95% CI [–.5, 1], $p = .35$, Yuen's t-test $t(64) = 1.7$, $p = .09$)



Note. Displayed are z-scores of cognitive performance at assessment two, as compared to baseline (line at zero). Point ranges denote mean and standard error of the mean (SEM).



Note. Displayed are z-scores of depression and fatigue at assessment two, as compared to baseline (line at zero). Point ranges denote mean and standard error of the mean (SEM).

Fatigue and Depression

- Patients reported **lower levels of depression** ($V = 719$, $HL = –1.5$, 95% CI [–2.5, –0.5], $p < .01$).
- Patients' reports of **fatigue severity tended to be lower** ($V = 866.5$, $HL = –2$, 95% CI [–3.5, <0], $p = .041$).

Abbreviations:
Q1, 3 = quartiles 1, 3;
HL = Hodges-Lehmann estimator, a robust measure of location (median of all possible pairwise differences between observations);
** $p < .01$, * $p < .05$

Conclusions

- The **profile of neurocognitive deficits remained stable**, in quality and quantity: no observed changes in any cognitive domain
- Lower levels of self-reported **symptoms of depression and fatigue severity**
- This follow-up after early first assessment offers evidence for the **stability of neurocognitive deficits** and their potentially **chronic course.**

Previous study:
Kozik, Reuken, Utech, et al., *Frontiers in Psychology* **2023**, 14, 1233144.
doi:10.3389/fpsyg.2023.1233144